

SSB "Single side Band"

$$S(t)_{\text{DSB-SC}} = A_c A_m \cos 2\pi f_m t \cos 2\pi f_c t.$$

USB

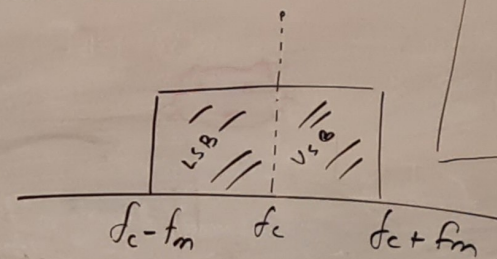
$$S(t)_{\text{DSB-SC}} = \frac{A_m A_c}{2} \cos(2\pi (f_c + f_m)t)$$

LSB

$$+ \frac{A_m A_c}{2} \cos(2\pi (f_c - f_m)t)$$

$$S(t)_{\text{SSB-SC}} = \frac{A_m A_c}{2} \cos(2\pi (f_c + f_m)t) \rightarrow \text{for upper side Band.}$$

$$S(t)_{\text{SSB-SC}} = \frac{A_m A_c}{2} \cos(2\pi (f_c - f_m)t) \rightarrow \text{for Lower side Band.}$$

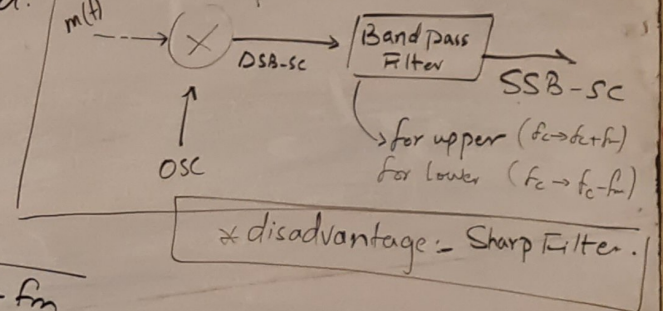


$$\bullet \text{ Bandwidth} = f_c + f_m - f_c = f_m$$

$$\bullet P_t = P_{\text{USB}} = P_{\text{LSB}} = \frac{A_m^2 A_c^2}{8R}$$

$$\bullet \eta = \frac{P_{\text{SB}}}{P_t} = 100\%$$

A. Frequency discrimination Modulation



$$S(t)_{DSB-SC} = A_c A_m \cos 2\pi f_m t \cos 2\pi f_c t.$$

USB

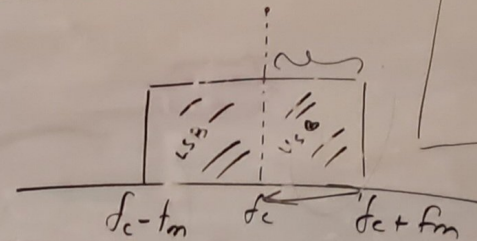
$$S(t)_{DSB-SC} = \frac{A_m A_c}{2} \cos(2\pi (f_c + f_m)t) + \frac{A_m A_c}{2} \cos(2\pi (f_c - f_m)t)$$

LSB.

$$S(t)_{SSB-SC} = \frac{A_m A_c}{2} \cos(2\pi (f_c + f_m)t) \rightarrow \text{for upper side Band}$$

$$S(t)_{SSB-SC} = \frac{A_m A_c}{2} \cos(2\pi (f_c - f_m)t) \rightarrow \text{for Lower side Band}$$

SSB "Single side Band"

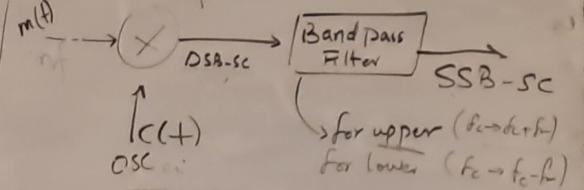


$$\text{Bandwidth} = f_c + f_m - f_c = f_m$$

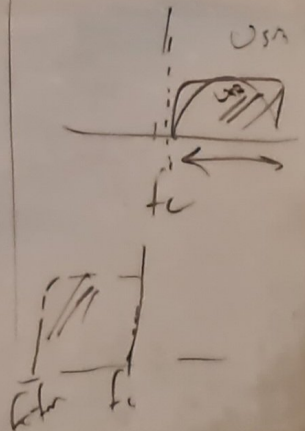
$$P_t = P_{USB} = P_{LSB} = \frac{A_m^2 A_c^2}{8R}$$

$$\eta = \frac{P_{SB}}{P_t} = 100\%$$

A. Frequency discrimination Modulation

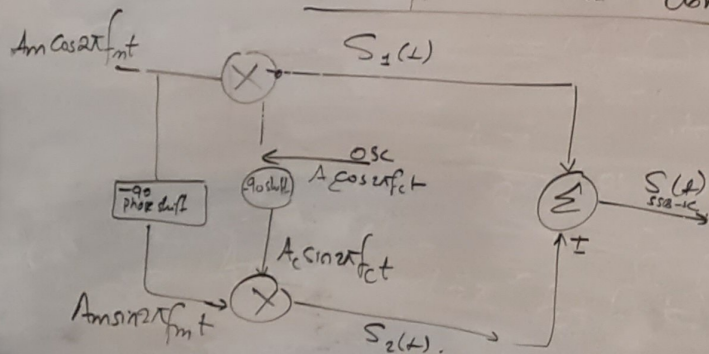


* disadvantage - Sharp Filter.



Phase discrimination Method.

*(Hilbert t) disadvantage
 * Difficult and Complex to implement.



$$S_1(t) = \frac{A_m A_c}{2} [\cos 2\pi (f_c + f_m)t + \cos 2\pi (f_c - f_m)t]$$

$$S_2(t) = \frac{A_m A_c}{2} [\cos 2\pi (f_c - f_m)t - \cos 2\pi (f_c + f_m)t]$$

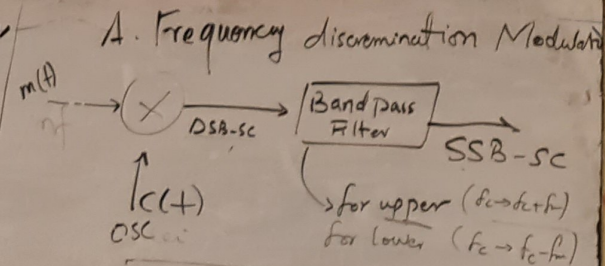
SSB "Single side Band"

Lower side (add)

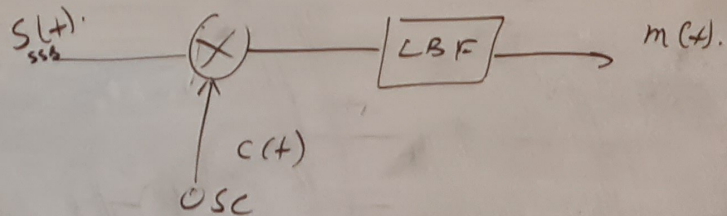
$$S(t)_{SSB-LSB} = \frac{A_m A_c}{2} [\cos 2\pi (f_c + f_m)t + \cos 2\pi (f_c - f_m)t] + \left[\frac{A_m A_c}{2} [\cos 2\pi (f_c - f_m)t - \cos 2\pi (f_c + f_m)t] \right]$$

$$S(t)_{SSB-LSB} = A_m A_c \cos 2\pi (f_c - f_m)t$$

For upper sideband (subtract)



Coherent detector.



$$s(t) \cdot c(t)$$

$$s(t) = \frac{A_m A_c}{2} \cos(2\pi(f_c - f_m)t) \cdot A_c \cos 2\pi f_c t$$

$$= \frac{A_m A_c^2}{2} \cos 2\pi f_m t + \frac{A_m A_c^2}{4} \cos(2\pi(2f_c - f_m)t)$$

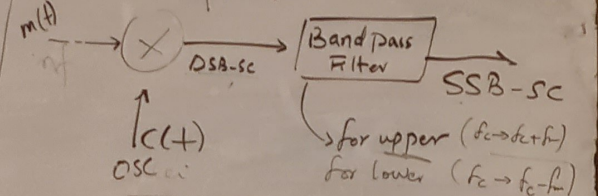
LBF.

$$v(t) = \frac{A_m A_c^2}{4} \cos(2\pi f_m t)$$

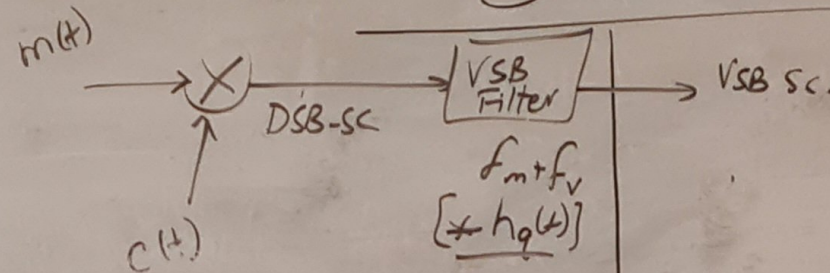
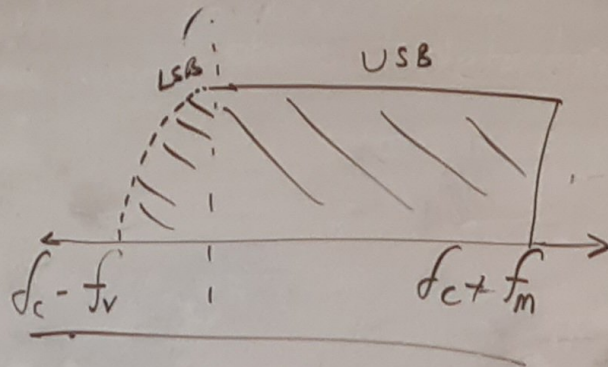
scaling factor $\left(\frac{A_c^2}{2}\right)$

SSB "Single side Band"

A. Frequency discrimination Modulation



SSB "Single side Band"



Bandwidth = $f_m + f_v$
 $\hookrightarrow$ vestige frequency.

$$f_v \approx 0.25 f_m$$

$$* \quad BW \leq f_m + 0.25 f_m = 1.25 f_m \text{ Hz.}$$

$$\rightarrow P_{VSB} = \alpha P_{LSB} + \beta P_{USB}$$

SSB "Single Side Band"

if $m(t) = 2\cos 2000\pi t + 4\cos 6000\pi t$.

and $c(t) = 10\cos 50 \times 10^3 \pi t$.

- Find:
- ① Mathematical Expression of SSB-USB/SSB-LSB
 - ② determine Amp. of each side bands.
 - ③ determine total Bandwidth.
 - ④ draw Frequency domain.

Given:

$$\begin{array}{l|l} f_{m1} = 1000, A_{m1} = 2V & f_c = 25 \text{ kHz} \\ f_{m2} = 3000, A_{m2} = 4V & A_c = 10V. \end{array}$$

$$S(t)_{LSB} = \frac{10\cos(24 \times 2\pi t)}{2} + \frac{20\cos(22 \times 2\pi t)}{2}$$

$$S(t)_{USB} = 10\cos(26 \times 2\pi t) + 20\cos(28 \times 2\pi t)$$

$$S(t)_{SSB-USB} = \left[\frac{A_m A_c}{2} \cos(2\pi(f_c + f_m)t) \right]$$

$$S(t)_{SSB-LSB} = \left[\frac{A_m A_c}{2} \cos(2\pi(f_c - f_m)t) \right]$$

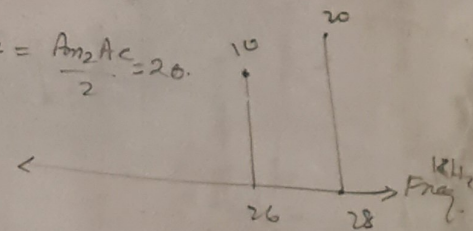
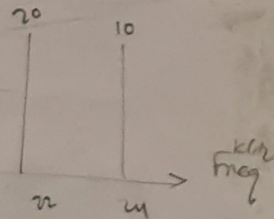
Amplitude of each side band = $\frac{A_m A_c}{2}$

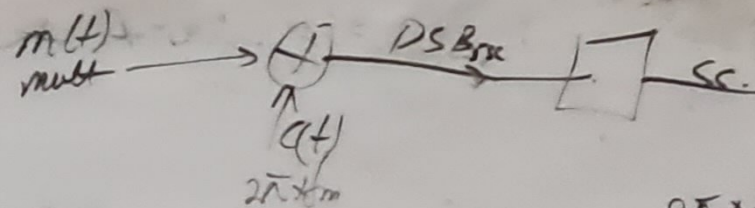
for $f_{m1} = \frac{A_{m1} A_c}{2} = 10$

for $f_{m2} = \frac{A_{m2} A_c}{2} = 20$

$B_{USB} =$

$B_{LSB} =$





SSB "Sing"

if $m(t) = \underbrace{2 \cos 2000\pi t}_{m_1(t)} + \underbrace{4 \cos 6000\pi t}_{m_2(t)}$

mod $c(t) = \underbrace{10 \cos 50 \times 10^3 \pi t}_{c(t)}$

- Find:
- ① Mathematical Expression of SSB-USB/SSB-LSB
 - ② determine Amp. of each side bands.
 - ③ determine total Bandwidth.
 - ④ draw Frequency domain.

$S(t)$
SSB-USB

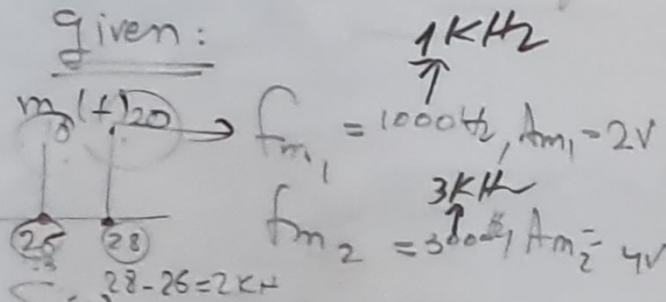
$S(t)$
SSB-LSB

• Amplitude each

• Bandwidth

Bandwidth

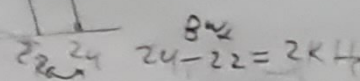
Given:



$f_c = 25 \text{ kHz}$
 $A_c = 10 \text{ V}$

$S(t)$
 $LSB(t) = 10 \cos(26 \times 2\pi t) + 20 \cos(28 \times 2\pi t)$

$S(t)$
 $USB = 10 \cos(24 \times 2\pi t) + 20 \cos(24 \times 2\pi t)$



SB "Single side Band"

$$S(t)_{SSB-USB} = \left[\frac{A_m A_c}{2} \cos(2\pi (f_c + f_m) t) \right]$$

$$S(t)_{SSB-LSB} = \left[\frac{A_m A_c}{2} \cos(2\pi (f_c - f_m) t) \right]$$

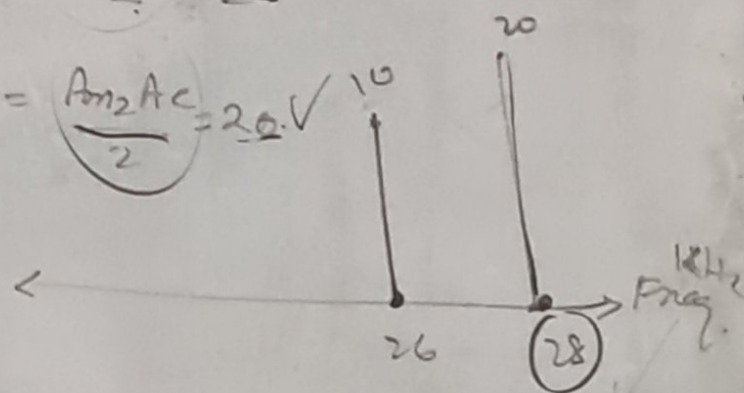
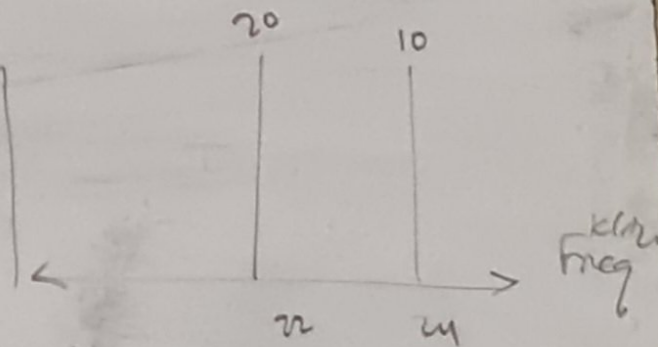
- Amplitude of each side band = $\frac{A_m A_c}{2}$

$$B_{W_{USB}} = f_{Max} - f_{Min}$$

$$B_{W_{LSB}} =$$

$$\rightarrow \text{for } f_{m1} = \frac{A_{m1} A_c}{2} = 10 \text{ V}$$

$$\rightarrow \text{for } f_{m2} = \frac{A_{m2} A_c}{2} = 20 \text{ V}$$



SSB "Single Side Band"

a base band message $m(t) = \cos(\frac{2000\pi t}{w}) + \cos(\frac{8000\pi t}{w})$ is modulate with $c(t) = 10 \cos(\frac{200 \times 10^3 \pi t}{w})$.

The VSB shaping Filter has a Following design:

- Transmitte full USB
- Vestigial LSB with $f_v = 2 \text{ kHz}$

Find: ① mathematical Expression of DSB-SC, ② Analysis the effect of VSB filter on the side bands, ③ Find mathematical expression of (VSB) modulated signal.

Solution: $f_1 = 1000 \text{ Hz} = 1 \text{ kHz}$, $f_2 = 4000 \text{ Hz} = 4 \text{ kHz}$, $f_c = 100 \text{ kHz}$, $A_c = 10 \text{ V}$, $A_{m_{1,2}} = 1 \text{ V}$.

① $S(t)_{\text{DSB-SC}} = m(t) \cdot c(t) = [\cos 2000\pi t + \cos 8000\pi t] \cdot [10 \cos 200 \times 10^3 \pi t]$ → Note that $\cos A + \cos B = \frac{1}{2} (\cos(A+B) + \cos(A-B))$

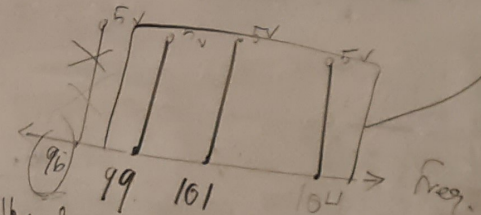
② $S(t)_{\text{DSB-S}} = \frac{5}{w} \cos(\frac{198000\pi t}{w}) + 5 \cos(\frac{202000\pi t}{w}) + 5 \cos(\frac{192000\pi t}{w}) + 5 \cos(\frac{208000\pi t}{w})$

$f_1 \ 99 \quad f_2 \ 101 \quad f_c \ 96 \quad f_c \ 104$

$\therefore f_c - f_v = 100 \text{ k} - 2 \text{ k} = 98 \text{ kHz}$

③ $S(t)_{\text{VSB-SC}} = 5 \cos(198000) + 5(202) + 5 \cos 202\pi t$

$f_c - f_v$ is neglected / filtered
 $96 < f_c - f_v = 98$



$$P = 1x$$

1000 Hz

SC R "Side Band"

look 10

total power 200 watt.

→ filter remove 80% of USB - & +x Full USB

↓
20% P_{USB} + 1 P_{USB}